

MOSKVA VNUKOVO, Russian Federation

WMO: 275185

Lat: 55.592N

Lon: 37.261E

Elev: 685

StdP: 14.34

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.4	-3.9	-14.8	2.5	-8.5	-9.1	3.5	-3.2	23.1	26.3	20.9	24.3	6.3	50	0.610

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.1	84.6	66.3	82.1	65.7	78.8	64.6	69.3	79.9	67.5	77.7	66.0	75.9	8.3	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.0	98.2	74.2	64.3	92.4	72.2	62.5	86.8	70.8	33.7	80.2	32.4	77.5	31.1	75.9	75.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	20.6	18.3	16.4	DB	-14.7	88.7	6.4	3.8	-19.3	91.4	-23.0	93.6	-26.6	95.8	-31.2	98.5
(5)				WB	-15.1	72.1	6.3	2.0	-19.6	73.6	-23.3	74.8	-26.8	76.0	-31.4	77.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	42.3	18.5	19.4	28.9	43.7	55.5	62.0	66.8	63.3	53.2	41.7	30.3	22.5	(6)
(7)		DBStd	19.29	11.16	11.54	8.33	8.85	8.81	7.12	6.32	6.83	7.21	7.74	9.24	11.04	(7)
(8)		HDD50	4536	975	858	656	230	47	3	0	0	47	275	592	853	(8)
(9)		HDD65	8538	1440	1278	1121	639	313	140	54	113	357	723	1042	1318	(9)
(10)		CDD50	1718	0	0	0	41	217	363	522	414	144	17	0	0	(10)
(11)		CDD65	244	0	0	0	0	18	50	111	61	4	0	0	0	(11)
(12)		CDH74	2032	0	0	0	4	163	433	878	525	29	0	0	0	(12)
(13)		CDH80	514	0	0	0	0	27	91	246	149	1	0	0	0	(13)
(14)	Wind	WSAvg	8.3	8.8	9.0	9.0	8.3	8.1	7.8	7.2	7.3	7.4	8.6	8.8	9.2	(14)
(15)	Precipitation	PrecAvg	26.2	1.9	1.5	1.4	1.3	2.2	2.8	3.1	2.9	2.6	2.6	2.0	1.8	(15)
(16)		PrecMax	34.6	3.3	3.1	2.9	2.2	4.6	4.6	6.6	5.7	7.3	5.3	3.4	3.1	(16)
(17)		PrecMin	17.2	0.6	0.3	0.6	0.5	0.7	0.4	0.5	0.5	0.5	0.6	0.1	0.6	(17)
(18)		PrecStd	4.3	0.7	0.6	0.6	0.5	1.0	1.0	1.6	1.5	1.3	1.4	0.7	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.0	38.7	55.3	73.4	82.8	86.1	91.4	91.1	77.4	65.9	52.0	44.8	(19)
(20)		MCWB	37.5	36.4	44.9	56.2	64.6	66.4	68.5	66.7	63.0	54.8	48.7	43.8	(20)	
(21)		2%	DB	35.7	36.0	48.1	68.0	78.8	82.5	85.9	83.9	73.2	60.4	46.8	40.8	(21)
(22)		MCWB	34.8	34.3	39.8	53.1	62.1	65.7	66.8	66.3	60.3	53.4	45.3	39.7	(22)	
(23)		5%	DB	33.9	35.2	43.0	62.9	73.8	79.0	82.3	79.1	68.4	55.6	44.5	37.3	(23)
(24)		MCWB	33.3	33.7	37.3	50.6	59.1	64.0	66.1	65.1	58.9	51.0	43.0	36.4	(24)	
(25)		10%	DB	32.2	33.7	39.4	58.8	70.1	75.3	78.8	75.5	64.5	52.1	41.2	34.1	(25)
(26)		MCWB	31.6	32.7	35.6	48.2	57.1	62.1	65.5	63.8	56.7	48.6	40.0	33.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.3	36.9	44.9	57.7	67.0	70.8	72.4	70.8	65.3	57.5	49.7	43.9	(27)
(28)		MCDWB	37.9	37.5	53.1	70.8	80.1	81.4	82.9	81.0	73.6	62.2	52.2	44.4	(28)	
(29)		2%	WB	34.4	34.8	40.8	54.3	64.0	67.7	69.8	68.2	61.9	54.4	45.5	39.1	(29)
(30)		MCDWB	34.8	35.4	45.6	64.5	75.6	78.7	80.0	78.5	69.9	58.9	46.5	40.0	(30)	
(31)		5%	WB	33.0	33.5	38.0	51.9	61.4	65.6	68.0	66.5	59.6	51.7	42.6	35.9	(31)
(32)		MCDWB	33.4	34.1	41.4	61.2	71.4	75.7	77.8	77.0	66.7	55.9	43.5	36.5	(32)	
(33)		10%	WB	31.4	32.3	35.9	49.5	58.9	63.4	66.6	64.7	57.7	49.5	39.8	33.3	(33)
(34)		MCDWB	31.9	33.1	38.9	57.3	67.1	72.6	76.1	73.9	64.1	51.8	40.6	33.7	(34)	
(35)	Mean Daily Temperature Range	MDBR	8.5	10.5	12.5	16.0	17.9	17.4	17.1	17.0	14.4	10.3	7.3	7.6	(35)	
(36)		5% DB	MCDBR	6.7	8.1	17.5	22.3	22.4	22.3	21.8	21.8	20.3	15.8	10.1	6.6	(36)
(37)		MCWBR	6.7	7.4	11.6	12.1	10.7	10.0	8.3	9.3	10.7	10.5	8.9	6.5	(37)	
(38)		5% WB	MCDBR	6.5	7.1	14.0	20.2	20.4	20.0	18.6	19.2	18.0	14.3	8.9	6.4	(38)
(39)		MCWBR	6.7	7.1	9.8	11.8	10.7	10.6	8.9	9.7	10.4	10.1	8.5	6.4	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.291	0.330	0.347	0.423	0.402	0.388	0.435	0.440	0.390	0.357	0.322	0.287	(40)	
(41)		taud	2.120	2.058	2.116	2.108	2.289	2.351	2.243	2.209	2.328	2.356	2.236	2.120	(41)	
(42)		Ebn at Noon	190	224	255	251	266	272	255	244	241	220	179	160	(42)	
(43)		Edh at Noon	20	31	37	43	38	36	40	39	30	23	18	16	(43)	
(44)	All-Sky Solar Radiation	RadAvg	190	460	948	1270	1666	1740	1704	1383	884	404	156	102	(44)	
(45)		RadStd	38	62	94	107	156	168	184	148	115	70	29	30	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.40	N/A	N/A	N/A	N/A	-0.86	-297	-410	+173	N/A
(47)	Regional (10 neighbors)	+1.33	N/A	N/A	N/A	N/A	N/A	-284	-410	+203	N/A

Nomenclature: See separate page